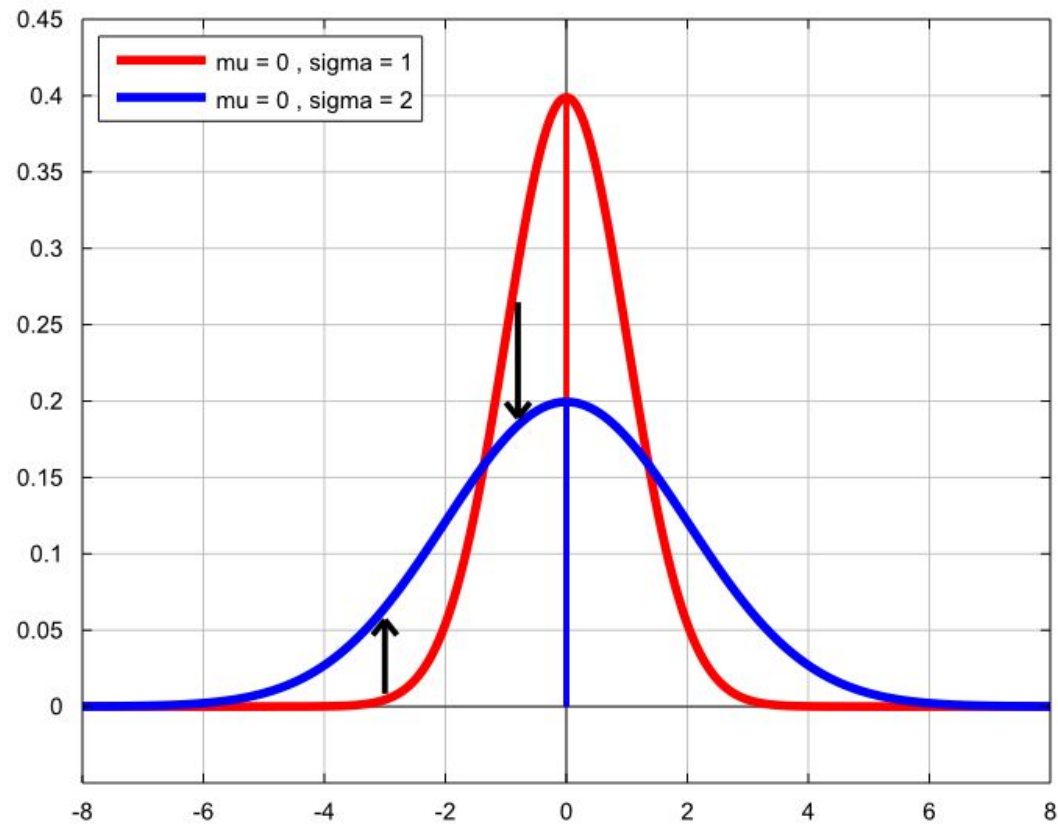


Gaussian Mixture Model for Clustering

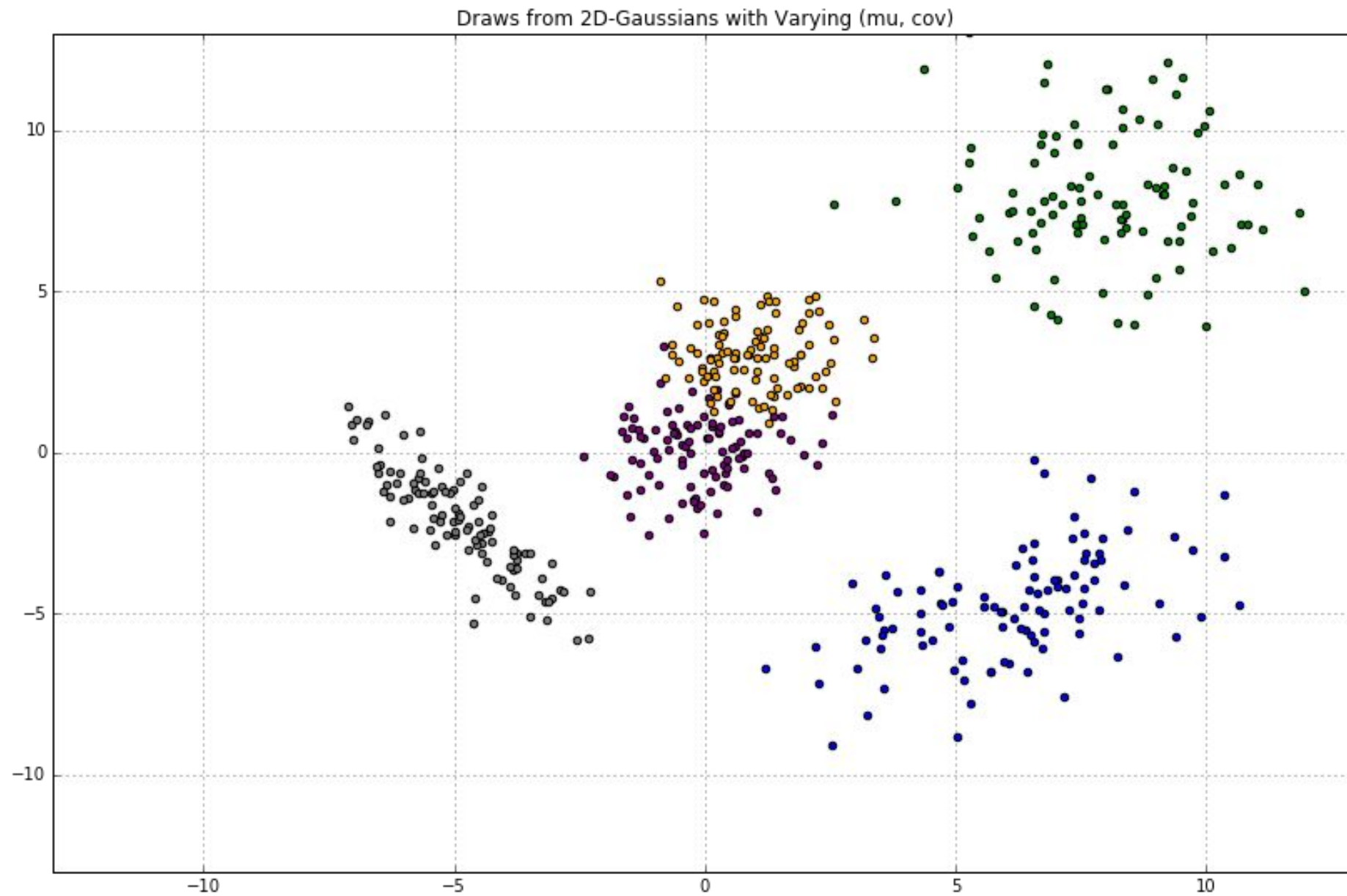
What EM can do in unsupervised learning?

Gaussian mixture models can be used to cluster unlabeled data in much the same way as k-means. There are, however, a couple of advantages to using Gaussian mixture models over k-means.

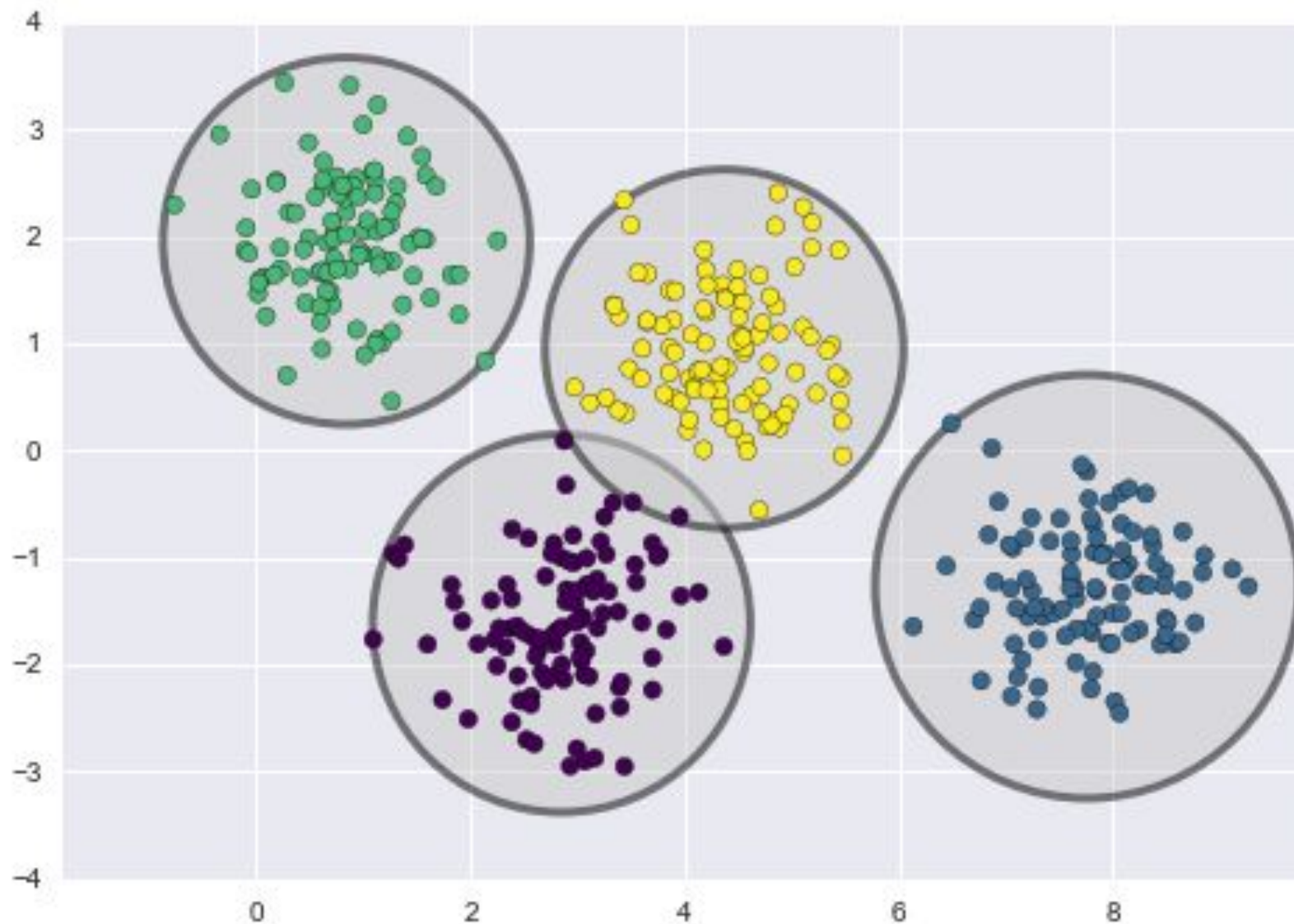
First and foremost, k-means does not account for variance. By variance, we are referring to the width of the bell shape curve.



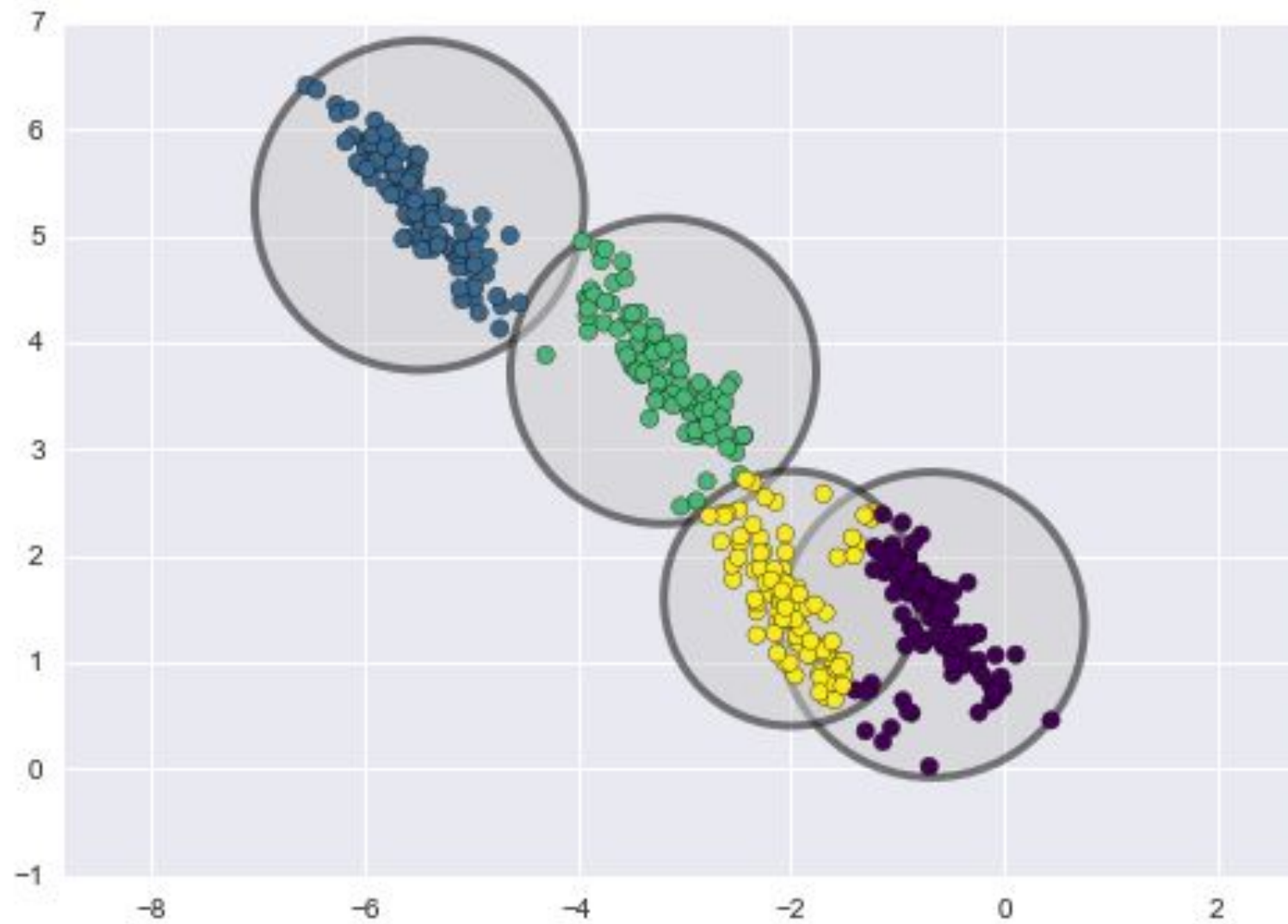
In two dimensions, variance (covariance to be exact) determines the shape of the distribution.



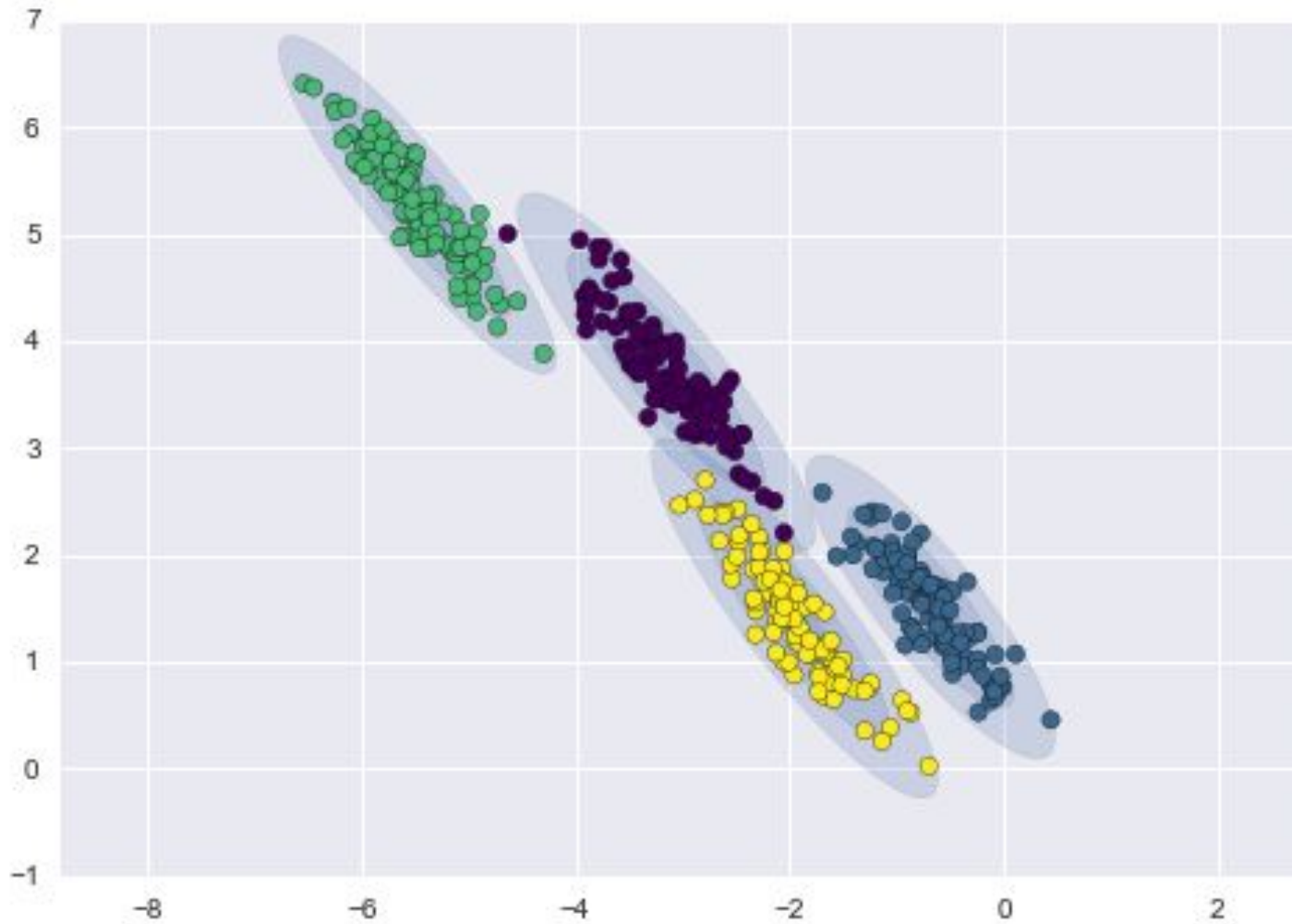
One way to think about the k -means model is that it places a circle (or, in higher dimensions, a hyper-sphere) at the center of each cluster, with a radius defined by the most distant point in the cluster.



This works fine for when your data is circular. However, when your data takes on different shape, you end up with something like this.



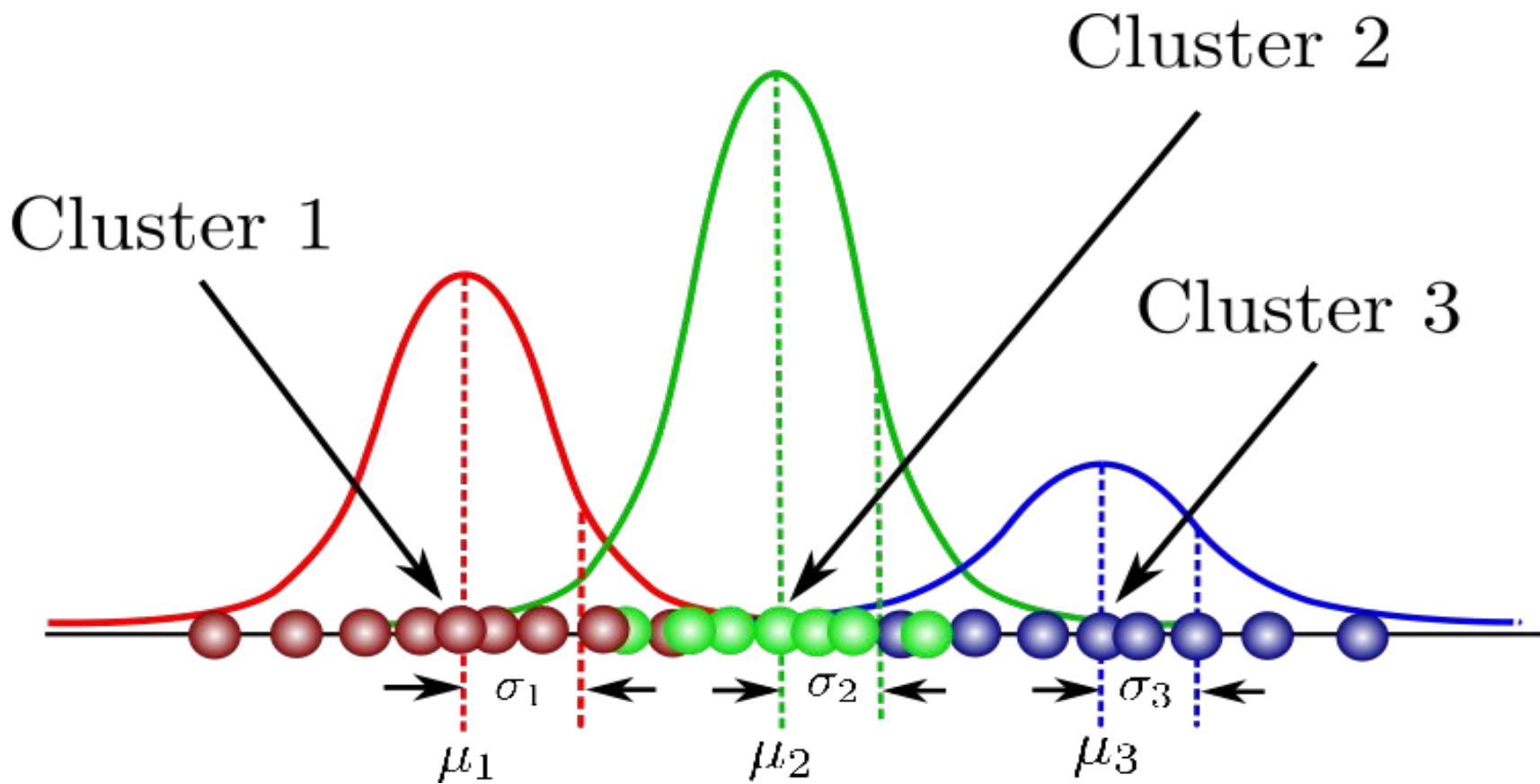
In contrast, Gaussian mixture models can handle even very oblong clusters.



The second difference between k-means and Gaussian mixture models is that the former performs hard classification whereas the latter performs soft classification. In other words, k-means tells us what data point belong to which cluster but won't provide us with the probabilities that a given data point belongs to each of the possible clusters.

A *Gaussian Mixture* is a function that is comprised of several Gaussians, each identified by $k \in \{1, \dots, K\}$, where K is the number of clusters of our dataset. Each Gaussian k in the mixture is comprised of the following parameters:

- A mean μ that defines its center.
- A covariance Σ that defines its width. This would be equivalent to the dimensions of an ellipsoid in a multivariate scenario.
- A mixing probability π that defines how big or small the Gaussian function will be.



Here, we can see that there are three Gaussian functions, hence $K = 3$. Each Gaussian explains the data contained in each of the three clusters available. The mixing coefficients are themselves probabilities and must meet this condition:

$$\sum_{k=1}^K \pi_k = 1 \quad (1)$$

Now how do we determine the optimal values for these parameters? To achieve this we must ensure that each Gaussian fits the data points belonging to each cluster. This is exactly what maximum likelihood does.

In general, the Gaussian density function is given by:

$$\mathcal{N}(\mathbf{x}|\mu, \Sigma) = \frac{1}{(2\pi)^{D/2}|\Sigma|^{1/2}} \exp \left(-\frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu) \right)$$

If we have a dataset comprised of $N = 1000$ three-dimensional points ($D = 3$), then $\mathbf{x}$ will be a 1000×3 matrix. μ will be a 1×3 vector, and Σ will be a 3×3 matrix. For later purposes, we will also find it useful to take the log of this equation, which is given by:

$$\ln \mathcal{N}(\mathbf{x}|\mu, \Sigma) = -\frac{D}{2} \ln 2\pi - \frac{1}{2} \ln |\Sigma| - \frac{1}{2}(\mathbf{x} - \mu)^T \Sigma^{-1}(\mathbf{x} - \mu) \quad (2)$$

If we differentiate this equation with respect to the mean and covariance and then equate it to zero, we will be able to find the optimal values for these parameters, and the solutions will correspond to the Maximum Likelihood Estimates (MLE) for this setting.

Suppose we want to know what is the probability that a data point $\mathbf{x}_n$ comes from Gaussian k . We can express this as:

$$p(z_{nk} = 1 | \mathbf{x}_n)$$

which reads “*given a data point $\mathbf{x}$, what is the probability it came from Gaussian k ?*” In this case, z is a *latent variable* that takes only two possible values. It is one when $\mathbf{x}$ came from Gaussian k , and zero otherwise.

Likewise, we can state the following:

$$\pi_k = p(z_k = 1)$$

Which means that the overall probability of observing a point that comes from Gaussian k is actually equivalent to the mixing coefficient for that Gaussian. This makes sense, because the bigger the Gaussian is, the higher we would expect this probability to be.

Now let $\mathbf{z}$ be the set of all possible latent variables z , hence:

$$\mathbf{z} = \{z_1, \dots, z_K\}$$

We know beforehand that each z occurs independently of others and that they can only take the value of one when k is equal to the cluster the point comes from. Therefore:

$$p(\mathbf{z}) = p(z_1 = 1)^{z_1} p(z_2 = 1)^{z_2} \dots p(z_K = 1)^{z_K} = \prod_{k=1}^K \pi_k^{z_k}$$

Now, what about finding the probability of observing our data given that it came from Gaussian k ? Turns out to be that it is actually the Gaussian function itself! Following the same logic we used to define $p(\mathbf{z})$, we can state:

$$p(\mathbf{x}_n | \mathbf{z}) = \prod_{k=1}^K \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)^{z_k}$$

Ok, now you may be asking, why are we doing all this? Remember our initial aim was to determine what the probability of z given our observation x ? Well, it turns out to be that the equations we have just derived, along with the Bayes rule, will help us determine this probability.

From the product rule of probabilities, we know that

$$p(\mathbf{x}_n, \mathbf{z}) = p(\mathbf{x}_n | \mathbf{z})p(\mathbf{z})$$

So how do we get rid of $\mathbf{z}$ here? Yes, you guessed it right. **Marginalization**! We just need to sum up the terms on $\mathbf{z}$, hence

$$p(\mathbf{x}_n) = \sum_{k=1}^K p(\mathbf{x}_n | \mathbf{z})p(\mathbf{z}) = \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)$$

This is the equation that defines a Gaussian Mixture, and you can clearly see that it depends on all parameters that we mentioned previously!

We can find the likelihood as the joint probability of all observations $\mathbf{x}_n$, defined by:

$$p(\mathbf{X}) = \prod_{n=1}^N p(\mathbf{x}_n) = \prod_{n=1}^N \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)$$

As usual, then by taking the log-likelihood:

$$\ln p(\mathbf{X}) = \sum_{n=1}^N \ln \sum_{k=1}^K \pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k) \quad (3)$$

Great! Now in order to find the optimal parameters for the Gaussian mixture, all we have to do is to differentiate this equation with respect to the parameters and we are done, right?

Wait! Not so fast.

We have an issue here. We can see that there is a logarithm that is affecting the second summation. Calculating the derivative of this expression and then solving for the parameters is going to be very hard!

Remember we were supposed to find the probability of z given $\mathbf{x}$?

From Bayes rule, we know that

$$p(z_k = 1 | \mathbf{x}_n) = \frac{p(\mathbf{x}_n | z_k = 1)p(z_k = 1)}{\sum_{j=1}^K p(\mathbf{x}_n | z_j = 1)p(z_j = 1)}$$

From our earlier derivations we learned that:

$$p(z_k = 1) = \pi_k, \quad p(\mathbf{x}_n | z_k = 1) = \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)$$

So let's now replace these in the previous equation:

$$p(z_k = 1 | \mathbf{x}_n) = \frac{\pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_n | \mu_j, \Sigma_j)} = \gamma(z_{nk}) \quad (4)$$

And thereby comes the **Expectation Maximization (EM)** Algorithm:

Why EM?

Consider a 1-D Gaussian for simplicity....

Now we attempt the same strategy for deriving the MLE of the Gaussian mixture model. Our unknown parameters are $\theta = \{\mu_1, \dots, \mu_K, \sigma_1, \dots, \sigma_K, \pi_1, \dots, \pi_K\}$, and so from the first section of this note, our likelihood is:

$$L(\theta|X_1, \dots, X_n) = \prod_{i=1}^n \sum_{k=1}^K \pi_k N(x_i; \mu_k, \sigma_k^2)$$

So our log-likelihood is:

$$\ell(\theta) = \sum_{i=1}^n \log \left(\sum_{k=1}^K \pi_k N(x_i; \mu_k, \sigma_k^2) \right)$$

$$\ell(\theta) = \sum_{i=1}^n \log \left(\sum_{k=1}^K \pi_k N(x_i; \mu_k, \sigma_k^2) \right)$$

Taking a look at the expression above, we already see a difference between this scenario and the simple setup in the previous section. We see that the summation over the K components “blocks” our log function from being applied to the normal densities. If we were to follow the same steps as above and differentiate with respect to μ_k and set the expression equal to zero, we would get:

$$\sum_{i=1}^n \frac{1}{\sum_{k=1}^K \pi_k N(x_i; \mu_k, \sigma_k)} \pi_k N(x_i; \mu_k, \sigma_k) \frac{(x_i - \mu_k)}{\sigma_k^2} = 0 \quad (1)$$

Now we’re stuck because we can’t analytically solve for μ_k . However, we make one important observation which provides intuition for what’s to come: if we knew the latent variables Z_i , then we could simply gather all our samples X_i such that $Z_i = k$ and simply use the estimate from the previous section to estimate μ_k .

EM, informally

Intuitively, the latent variables Z_i should help us find the MLEs. We first attempt to compute the posterior distribution of Z_i given the observations:

$$P(Z_i = k | X_i) = \frac{P(X_i | Z_i = k)P(Z_i = k)}{P(X_i)} = \frac{\pi_k N(\mu_k, \sigma_k^2)}{\sum_{k=1}^K \pi_k N(\mu_k, \sigma_k^2)} = \gamma_{Z_i}(k) \quad (2)$$

Now we can rewrite equation (1), the derivative of the log-likelihood with respect to μ_k , as follows:

$$\sum_{i=1}^n \gamma_{Z_i}(k) \frac{(x_i - \mu_k)}{\sigma_k^2} = 0$$

Even though $\gamma_{Z_i}(k)$ depends on μ_k , we can cheat a bit and pretend that it doesn't. Now we can solve for μ_k in this equation to get:

$$\hat{\mu}_k = \frac{\sum_{i=1}^n \gamma_{z_i}(k) x_i}{\sum_{i=1}^n \gamma_{z_i}(k)} = \frac{1}{N_k} \sum_{i=1}^n \gamma_{z_i}(k) x_i \quad (3)$$

Where we set $N_k = \sum_{i=1}^n \gamma_{z_i}(k)$. We can think of N_k as the effective number of points assigned to component k . We see that $\hat{\mu}_k$ is therefore a weighted average of the data with weights $\gamma_{z_i}(k)$. Similarly, if we apply a similar method to finding $\hat{\sigma}_k^2$ and $\hat{\pi}_k$, we find that:

$$\hat{\sigma}_k^2 = \frac{1}{N_k} \sum_{i=1}^n \gamma_{z_i}(k) (x_i - \mu_k)^2 \quad (4)$$

$$\hat{\pi}_k = \frac{N_k}{n} \quad (5)$$

Again, remember that $\gamma_{z_i}(k)$ depends on the unknown parameters, so these equations are not closed-form expressions. This looks like a vicious circle. But, as Cosma Shalizi (https://en.wikipedia.org/wiki/Cosma_Shalizi) says, “one man’s vicious circle is another man’s successive approximation procedure.”

We are now in the following situation:

1. If we knew the parameters, we could compute the posterior probabilities $\gamma_{z_i}(k)$
2. If we knew the posteriors $\gamma_{z_i}(k)$, we could easily compute the parameters

The EM algorithm, motivated by the two observations above, proceeds as follows:

1. Initialize the μ_k 's, σ_k 's and π_k 's and evaluate the log-likelihood with these parameters.
2. **E-step:** Evaluate the posterior probabilities $\gamma_{Z_i}(k)$ using the current values of the μ_k 's and σ_k 's with equation (2)
3. **M-step:** Estimate new parameters $\hat{\mu}_k, \hat{\sigma}_k^2$ and $\hat{\pi}_k$ with the current values of $\gamma_{Z_i}(k)$ using equations (3), (4) and (5).
4. Evaluate the log-likelihood with the new parameter estimates. If the log-likelihood has changed by less than some small ϵ , stop. Otherwise, go back to step 2.

The EM algorithm is sensitive to the initial values of the parameters, so care must be taken in the first step. However, assuming the initial values are “valid,” one property of the EM algorithm is that the log-likelihood increases at every step. This invariant proves to be useful when debugging the algorithm in practice.

EM, formally

The EM algorithm attempts to find maximum likelihood estimates for models with latent variables. In this section, we describe a more abstract view of EM which can be extended to other latent variable models.

Let X be the entire set of observed variables and Z the entire set of latent variables. The log-likelihood is therefore:

$$\log(P(X|\Theta)) = \log\left(\sum_Z P(X, Z|\Theta)\right)$$

where we've simply marginalized Z out of the joint distribution.

As we noted above, the existence of the sum inside the logarithm prevents us from applying the log to the densities which results in a complicated expression for the MLE. Now suppose that we observed both X and Z . We call $\{X, Z\}$ the **complete** data set, and we say X is **incomplete**. As we noted previously, if we knew Z , the maximization would be easy.

We typically don't know Z , but the information we do have about Z is contained in the posterior $P(Z|X, \Theta)$. Since we don't know the complete log-likelihood, we consider its expectation under the posterior distribution of the latent variables. This corresponds to the **E-step** above. In the **M-step**, we maximize this expectation to find a new estimate for the parameters.

In the **E-step**, we use the current value of the parameters θ^0 to find the posterior distribution of the latent variables given by $P(Z|X, \theta^0)$. This corresponds to the $\gamma_{Z_i}(k)$ in the previous section. We then use this to find the expectation of the complete data log-likelihood, with respect to this posterior, evaluated at an arbitrary θ . This expectation is denoted $Q(\theta, \theta^0)$ and it equals:

$$Q(\theta, \theta^0) = E_{Z|X, \theta^0} [\log(P(X, Z|\theta))] = \sum_Z P(Z|X, \theta^0) \log(P(X, Z|\theta))$$

In the M-step, we determine the new parameter $\hat{\theta}$ by maximizing Q :

$$\hat{\theta} = \operatorname{argmax}_{\theta} Q(\theta, \theta^0)$$

Let the parameters of our model be

$$\theta = \{\pi, \mu, \Sigma\}$$

Let us now define the steps that the general EM algorithm will follow.

Step 1: Initialise θ accordingly. For instance, we can use the results obtained by a previous K-Means run as a good starting point for our algorithm.

Step 2 (Expectation step): Evaluate

$$Q(\theta^*, \theta) = \mathbb{E}[\ln p(\mathbf{X}, \mathbf{Z}|\theta^*)] = \sum_{\mathbf{Z}} p(\mathbf{Z}|\mathbf{X}, \theta) \ln p(\mathbf{X}, \mathbf{Z}|\theta^*) \quad (5)$$

Well, actually we have already found $p(\mathbf{Z}|\mathbf{X}, \theta)$. Remember the γ expression we ended up with in the previous section?

For better visibility, let's bring our earlier equation (4) here:

$$p(z_k = 1 | \mathbf{x}_n) = \frac{\pi_k \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)}{\sum_{j=1}^K \pi_j \mathcal{N}(\mathbf{x}_n | \mu_j, \Sigma_j)} = \gamma(z_{nk}) \quad (4)$$

For Gaussian Mixture Models, the expectation step boils down to calculating the value of γ in (4) by using the old parameter values. Now if we replace (4) in (5), we will have:

$$\mathcal{Q}(\theta^*, \theta) = \sum_{\mathbf{Z}} \gamma(z_{nk}) \ln p(\mathbf{X}, \mathbf{Z} | \theta^*) \quad (6)$$

Sounds good, but we are still missing $p(\mathbf{X}, \mathbf{Z} | \theta^*)$. How can we find it? Well, actually it's not that difficult. It is just the complete likelihood of the model, including both $\mathbf{X}$ and $\mathbf{Z}$, and we can find it by using the following expression:

$$p(\mathbf{X}, \mathbf{Z} | \theta^*) = \prod_{n=1}^N \prod_{k=1}^K \pi^{z_{nk}} \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)^{z_{nk}}$$

...and that is the result of calculating the joint probability of all observations and latent variables and is an extension of our initial derivations for $p(\mathbf{x})$. The log of this expression is given by

$$\ln p(\mathbf{X}, \mathbf{Z} | \theta^*) = \sum_{n=1}^N \sum_{k=1}^K z_{nk} [\ln \pi_k + \ln \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)] \quad (7)$$

Remember that the latent variable z will **only** be 1 once every time the summation is evaluated. With that knowledge, we can easily get rid of it as needed for our derivations.

Finally, we can replace (7) in (6) to get:

$$\mathcal{Q}(\theta^*, \theta) = \sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk}) [\ln \pi_k + \ln \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)] \quad (8)$$

In the maximization step, we will find the revised parameters of the mixture. For this purpose, we will need to make Q a restricted maximization problem and thus we will add a Lagrange multiplier to (8). Let's now review the maximization step.

Step 3 (Maximization step): Find the revised parameters θ^* using:

$$\theta^* = \arg \max_{\theta} Q(\theta^*, \theta)$$

Where

$$Q(\theta^*, \theta) = \sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk}) [\ln \pi_k + \ln \mathcal{N}(\mathbf{x}_n | \mu_k, \Sigma_k)] \quad (8)$$

However, Q should also take into account the restriction that all π values should sum up to one. To do so, we will need to add a suitable Lagrange multiplier. Therefore, we should rewrite (8) in this way:

$$Q(\theta^*, \theta) = \sum_{n=1}^N \sum_{k=1}^K \gamma(z_{nk}) [\ln \pi_k + \ln \mathcal{N}(x_n | \mu_k, \Sigma_k)] - \lambda \left(\sum_{k=1}^K \pi_k - 1 \right) \quad (8)$$

And now we can easily determine the parameters by using maximum likelihood. Let's now take the derivative of Q with respect to π and set it equal to zero:

$$\frac{\partial Q(\theta^*, \theta)}{\partial \pi_k} = \sum_{n=1}^N \frac{\gamma(z_{nk})}{\pi_k} - \lambda = 0$$

Then, by rearranging the terms and applying a summation over k to both sides of the equation, we obtain:

$$\sum_{n=1}^N \gamma(z_{nk}) = \pi_k \lambda \implies \sum_{k=1}^K \sum_{n=1}^N \gamma(z_{nk}) = \sum_{k=1}^K \pi_k \lambda$$

From (1), we know that the summation of all mixing coefficients π equals one. In addition, we know that summing up the probabilities γ over k will also give us 1. Thus we get $\lambda = N$. Using this result, we can solve for π :

$$\pi_k = \frac{\sum_{n=1}^N \gamma(z_{nk})}{N}$$

Similarly, if we differentiate Q with respect to μ and Σ , equate the derivative to zero and then solve for the parameters by making use of the log-likelihood equation (2) we defined, we obtain:

$$\mu_k^* = \frac{\sum_{n=1}^N \gamma(z_{nk}) \mathbf{x}_n}{\sum_{n=1}^N \gamma(z_{nk})}, \quad \Sigma_k^* = \frac{\sum_{n=1}^N \gamma(z_{nk}) (\mathbf{x}_n - \mu_k) (\mathbf{x}_n - \mu_k)^T}{\sum_{n=1}^N \gamma(z_{nk})}$$

And that's it! Then we will use these revised values to determine γ in the next EM iteration and so on and so forth until we see some convergence in the likelihood value. We can use equation (3) to monitor the log-likelihood in each step and we are always guaranteed to reach a local maximum.